Sahaj Bhadja

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Summary

Software engineer with 2+ years building **backend systems**, cloud APIs, and data pipelines. Strong in **AWS**, **Python**, **Node.js**-adjacent web APIs, SQL, Docker, and monitoring-backed workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
M.S. in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Tech, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
B.Tech. in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Eng., OS, OOP, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant platform services on Azure SQL and Blob Storage for 500+ users, optimizing queries to cut dashboard latency from 1.2s to 350ms in production.
- Built fault-tolerant async workflows for 1K+ AI jobs with state transitions and failure isolation, improving throughput 3x while supporting dependable backend releases and operational monitoring.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Flask services and REST APIs handling 10K+ daily requests, integrating secure file ingestion, grading pipelines, and production logic for AI-assisted application workflows.
- Designed concurrent batch-processing for 1K+ uploaded files using multithreading and AWS RDS-backed services, reducing turnaround time by 60% while maintaining stable APIs in production.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built spatial-temporal evaluation pipelines over 100K+ trajectory records, improving decision stability by 15% while strengthening reproducibility, model evaluation, and AI training workflows.
- Developed experimental tooling and reporting loops for sensor-driven systems, using iterative validation and failure analysis to communicate actionable findings across technical collaborators.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform software with microservices, Jenkins, PyTest, and Mockito, improving release reliability through 20+ automated test suites across Windows, Linux, and macOS.
- Optimized API and ETL workflows with AWS Glue, S3, and SQL-backed services, improving response times by 30% and pipeline efficiency by 68% for production operations.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed full-stack workflow software using REST APIs, WebSockets, and responsive interfaces, supporting 2M+ synchronized records while improving operational visibility, coordination, and system usability.
- Automated Docker-based build and deployment workflows with Jenkins CI/CD, cutting release time by 50% and improving troubleshooting speed, environment consistency, and delivery reliability.

Skills

Backend & APIs: Python, Flask, FastAPI, Node.js, REST APIs, microservices, WebSockets, SQL, PostgreSQL, MySQL, Azure SQL, local/server APIs
Cloud & Infra: AWS, AWS RDS, AWS Glue, S3, serverless concepts, Docker, Kubernetes, Linux, Bash, CI/CD, Jenkins, Git
Data, IoT & AI: ETL pipelines, data validation, monitoring workflows, metadata processing, model evaluation, MongoDB, MQTT familiarity, testing, debugging, security-minded design

Projects

Linguistix | *Python, AWS, Docker, CI/CD*
- Developed and deployed a containerized RESTful data processing API on AWS with CI/CD releases and validation checks, achieving 87% accuracy and reliable production behavior.

FinNexus Live | *Next.js, TypeScript, Plaid, MongoDB, Kubernetes*
- Built a scalable product supporting 50K+ users and 1M+ transactions, combining secure APIs, real-time updates, MongoDB-backed workflows, and Kubernetes-oriented deployment practices.

HMIS Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*
- Built a healthcare operations platform with scheduling, alerts, and dashboards, reducing administrative workload by 30% through structured TypeScript services, monitoring, and automation.